

# Cohort Reconstruction Sampling

Arthur Barros

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## Summary

Accurate estimates of age-specific natural-origin escapement are essential for cohort reconstruction (CR) of Sacramento River fall-run Chinook Salmon (SFCS). This methods document uses Monte Carlo simulations to address two sampling questions: (1) how many scales are needed to estimate natural-origin age-class proportions or abundances, and (2) whether pooling scales across the basin is preferable to sampling a fixed number from each location.

The simulations demonstrate that pooling scale samples with weighting by escapement contribution outperforms the alternative method of estimating age-class abundance for each location independently and summing results. The pooling method requires fewer total scales for aging, achieves higher coverage of true values, and exhibits lower bias. False negatives (failing to detect an age class when present) are rare, but false positives (estimating presence where none exists) occurs frequently due to uncertainty in hatchery abundance estimates. Based on the results of the Monte Carlo simulations I recommend adopting an approach of pooling scales to produce one basin-wide estimate, ensuring that scale sample sizes are allocated proportionally to each location’s contribution to total escapement.

## Introduction

Efforts to implement age-structured cohort reconstruction (CR) of Sacramento River fall-run Chinook Salmon (SFCS) populations are under way. The currently used Sacramento Index (SI) serves as an age-aggregated index of adult ocean abundance and potential adult escapement [1], and the CR aims to update this method by reporting age-specific assessments with estimates of uncertainty. Accurate estimates of age-specific escapement of the natural- and hatchery-origin components are necessary to accurately estimate the maturation rates, mortality rates, and rates of exploitation utilized in the CR. The hatchery-origin component of escapement is estimated using coded-wire-tag (CWT) rates, but the natural-origin component is reconstructed using scale age data collected from untagged fish. Scale aging can be labor-intensive and requires substantial time and training to be done accurately. This document aims to show how many scales should be aged to inform CR efforts in the Sacramento River basin and provide evidence to support the most efficient method of sampling scales.

Here I provide an overview of the methods used to estimate the number of scales required to accurately estimate the abundance of each age-class of natural-origin Sacramento River fall-run Chinook Salmon (SFCS). The analysis I review here is in response to two questions from the Scales/CWT subgroup supporting efforts to implement CR efforts in the Sacramento River basin but I also aim to expand upon those questions and give sampling suggestions.

My code scripts and documentation are written in the R language. Throughout this document, various chunks of R code will be presented in the following format:

```
print("hello reader")
```

```
[1] "hello reader"
```

These chunks are designed for users to copy and paste, or rewrite entirely, into their own R scripts to replicate the methods. Additionally, the code used in this document relies on the following R packages: `tidyverse`, `ggplot2`, `data.table`, `gtools`, and `gridExtra`. Code for all the functions, related data cleaning, and simulation run scripts can be found at [https://github.com/arthurbarros-CDFW/CR\\_sampling](https://github.com/arthurbarros-CDFW/CR_sampling).

## Sampling questions

At the December 2025 CR Scale Aging workshop, two questions were put to the group:

**Q1:** How many scales do we need to sample from recovered untagged fish to accurately estimate the proportion of age-classes in natural-origin fish in a given population?

**Q2:** Instead of aging some minimum number of scales from each tributary and hatchery, could we pool samples across the Sacramento River basin for a single CR? Could this be done to reduce effort/cost?

In addition to these two questions, I've also highlighted sub-questions that relate to the above:

- **Q1:** How many scale samples are needed to achieve a margin-of-error (MOE) in 95% of simulations/trials? MOE targets ensure sample sizes will inform CR estimates with reasonable confidence.
- **Q1:** How many are needed to minimize the number of false positives (estimating the presence of an age-class when none occurs) for a natural-origin age-class?
- **Q1:** How many are needed to minimize the number of false negatives (failing to estimate the presence of an age-class when they are available)?
- **Q2:** What are the assumptions of pooling samples across the basin for aging, and what could be lost with this approach?

## **Review of Cohort Reconstruction Methods**

Escapement to a given survey location includes marked and CWT tagged hatchery-origin fish, unmarked and (presumed) untagged hatchery-origin fish, and unmarked and untagged natural-origin fish. To estimate the abundances of each of these escapement populations at a given survey location Emily Chen et. al. [1] take the following steps:

1. First, for each CWT recovered, the estimate of the number of tagged fish present but unrecovered  $k$  is drawn from a negative binomial distribution  $k \sim NB(1, \theta)$  where  $\theta$  is the sampling fraction of the survey, or the recovery probability of the tag.
2. For each recovered tag of an age-class  $a$ ,  $k + 1$  is the estimated number of tagged fish present per tag. The estimate of total abundance of hatchery-origin fish with tags  $\hat{N}_{CWT,a}$  in the escapement population is:

$$\hat{N}_{CWT,a} = \sum_{t=1}^j (k_{t,a} + 1) \quad (1)$$

3. The total estimate of all hatchery-origin fish for an age-class in the escapement population ( $\hat{N}_{ho,a}$ ) is estimated by expanding  $\hat{N}_{CWT,a}$  by  $F_{prod}$ , the tagging rate of that release group.

4. The abundance of untagged fish in the population is estimated as:

$$\hat{N}_{NO-CWT} = \hat{N} - \hat{N}_{CWT} \quad (2)$$

5. Scale ages of unmarked fish recovered at the survey location are used to estimate the age composition of unmarked escapement. For each age-class  $a$ , we can:

- calculate the number of untagged fish scales from that scale sample ( $n_{NO-CWT,a}$ ).
- calculate the proportion of untagged fish scales from that cohort ( $p_{NO-CWT,a}$ ).
- estimate the total number of untagged fish in the population  $\hat{N}_{NO-CWT,a}$  from that cohort:

$$\hat{N}_{NO-CWT,a} = \hat{N}_{NO-CWT} * p_{NO-CWT,a} \quad (3)$$

6. The estimate of the number of untagged hatchery-origin fish is done as:

$$\hat{N}_{NO-CWT,ho,a} = \hat{N}_{ho,a} - \hat{N}_{CWT,a} \quad (4)$$

Finally, the estimate of the abundance of natural-origin fish at each age-class is the number of untagged hatchery-origin fish at each age-class subtracted from the total number of untagged fish in that age-class:

$$\hat{N}_{no,a} = \hat{N}_{NO-CWT,a} - \hat{N}_{NO-CWT,ho,a} \quad (5)$$

Chen et. al. then take these estimates of hatchery-origin fish  $\hat{N}_{ho,a}$  and natural-origin fish  $\hat{N}_{no,a}$  for each age-class at each survey location and sum them together to create Sacramento River basin estimates of escapement for each cohort:

$$\hat{N}_{basin,no,a} = \sum_{l=1}^j \hat{N}_{l,no,a} \quad (6)$$

The above CR methodology requires collecting scale samples from tributary spawning grounds across the Sacramento River basin incorporates uncertainty from several sources:

- Fish recovery rate (  $\theta$  ) - The sampling fraction of the survey, or probability of recovering the tag of a given fish. This is derived from the “estimated number” reported in RMIS.
- Tag rate expansion ( $F_{prod}$ ) - The tag rate expansion method used to estimate number of all hatchery-origin fish in the population (tagged and untagged).

- Scale subsampling ( $n_{NO-CWT}$ ) - The number of scales sampled from the untagged recovered fish for aging. Scales that are sampled can be bootstrapped by redrawing with replacement (typically 1000 iterations) to characterize sampling uncertainty.

There are two other sources of uncertainty that are either not dealt with in the CR methods, or not dealt with in my analysis:

- Escapement estimates ( $\hat{N}$ ) - Estimates of total SFCS in natural spawning grounds are reported as point estimates in GrandTab, but there is significant uncertainty in these estimates generated by the Cormack-Jolly-Seber mark-recapture models. This uncertainty is not incorporated in the current CR methods and is beyond the scope of this analysis, but could be expanded upon in the future.
- Scale aging bias and correction - The age distribution of spawners is corrected using a confusion matrix developed by comparing read age from scales from fish with known date of birth from CWT. The scale aging confusion matrix would be standard across methods and its omission doesn't affect relative comparisons across sampling stages.

The multiple steps and sources of uncertainty incorporated in the CR methods mean that there is no closed-form equation readily available to estimate the number of scale samples needed to accurately estimate the size of each age-class of natural-origin SFCS. To address these questions I use a series of Monte Carlo simulations that vary sample size, effort ( $\theta$ ), and sampling strategy (pooling and no-pooling) to replicate the proposed sampling methods and evaluate their performance.

## **Monte Carlo Simulations**

A Monte Carlo approach involves running iterative simulations in which a hypothetical “true” population is generated, samples are drawn from this population, and criteria such as bias, error, and coverage are produced and analysed [3, 2].

The Monte Carlo approach used here is the following:

1. Simulate known populations of SFCS for single survey locations (Approach 1) or multiple locations making up a larger basin (Approach 2).
2. Simulate recovery of tagged and untagged fish from surveys and generate estimates of hatchery-origin abundance using the  $k$ -draw methods to estimate uncertainty.
3. Replicate scale sampling and bootstrapping of scales to characterize uncertainty.
4. Run simulations across various parameters and scenarios.
5. Analyze bias, coverage, and margin-of-error criteria to determine target scale sample sizes.

The analysis proceeds in two stages. First, I use simple theoretical populations to understand the fundamental relationships between sample size, sampling effort, and estimate precision. Second, I use empirically-based simulations to evaluate basin-wide sampling strategies.

### **Approach 1: simulating hypothetical known populations.**

- This first approach simulates completely theoretical escapement populations for a single location, runs sampling efforts to simulate fish and scale recovery, and then estimates the abundance of natural-origin fish at each age-class in the total escapement population. The sampling estimates are then compared to the generated true values to test accuracy, error, and coverage. This method is then repeated across different scenarios to better understand how various parameters such as sample size and  $\theta$  impact the estimates.

### **Approach 2: simulating Sacramento River basin populations.**

- The second approach utilizes actual estimates of hatchery and natural-origin age-specific abundance for Sacramento River basin spawning grounds and hatcheries produced by Emily Chen [1] available on GitHub (<https://github.com/echenfishbitch/SRFC-cohort-reconstruction-wNF>). These estimates are used to simulate basin-wide populations, and then random sampling is performed replicating the methods in approach 1. I then compare the method of basin-wide population estimation used by Emily Chen (Equation 6) to the alternative method of pooling scales across survey locations and randomly sampling from the pool. This addresses the point brought up by sampling question 2.

All Monte Carlo simulations were run on a HP ProBook 640 G5 and took between 2 to 6 hours to process.

### ***Approach 1: simulating theoretical populations***

I have produced functions in R that run many of the steps involved in the Monte Carlo simulations. These functions are available on the project GitHub repository under the `scripts/functions/` folder.

### **Simulate a known population**

The first function simulates a theoretical SFCS population for sampling with user input parameters.

`sim_pop()` - simulates a theoretical population using specific user inputs for parameters:

- `N` = true abundance.
- `h_prop` = proportion of population of hatchery origin fish. Default 0.75.
- `tag_rate` = hatchery-origin fish tag rate ( $F_{prod}$ ). Default 0.25.

- `ages` = possible fish ages. Default ages 2, 3, and 4.
- `age_probs` = fixed proportions of each age-class. Default 0.15, 0.60, 0.25.
- `random_age_probs`=whether to add random “noise” to age probabilities. Default TRUE.
- `age_var_origin` = whether to use different probabilities for hatchery vs natural origin fish. Default TRUE.
- `age_dirichlet_concentration` = Dirichlet concentration value to use to add random noise to age probabilities. By default, `sim_pop()` introduces random variation in age composition between hatchery- and natural-origin fish using a Dirichlet distribution. Default 100.

We can run a simple call of `sim_pop()` and see the below output `sim1`: a simulated population of 10,000 returning SFCS, a data frame with each record representing a fish, it’s origin, age, and whether or not is has a CWT.

```
sim1<-sim_pop(N = 10000)
head(sim1)
```

|   | fish_id | origin   | age | has_cwt |
|---|---------|----------|-----|---------|
| 1 | 1       | hatchery | 4   | 0       |
| 2 | 2       | hatchery | 3   | 0       |
| 3 | 3       | hatchery | 3   | 1       |
| 4 | 4       | natural  | 4   | 0       |
| 5 | 5       | hatchery | 3   | 0       |
| 6 | 6       | hatchery | 3   | 0       |

### Simulate fish recovery

Now that we’ve generated a simulated population, we can simulate the recovery of fish for a survey effort. Over a given spawning period some portion of the escapement will be recovered, which can be characterized with  $\theta$ . This process is implemented in the `sim_spawning_recovery()` function and the following inputs:

- `sim_population` = the data frame of the simulated population.
- `theta` ( $\theta$ )= the probability of recovering a given fish, or the sampling fraction. Default 0.20.
- `tag_rate` = hatchery-origin fish tag rate ( $F_{prod}$ ). Default 0.25.
- `iterations` = number of draws from a negative-binomial distribution to estimate  $k$ , where  $k \sim NB(1, \theta)$ . Default 1000.

We can use the default values to simulate a spawning recovery event below.

```
spawning_recoveries <- sim_spawning_recovery(sim_population = sim1)
```

The `sim_spawning_recovery()` function outputs a list with two data frames, the first is `recovered_fish`, a table of all the fish “recovered” from the total population. This output gives the true age and origin for each fish and whether that fish has a CWT.

```
#show table of recovered fish
head(spawning_recoveries$recovered_fish)
```

|   | fish_id | origin   | age | has_cwt |
|---|---------|----------|-----|---------|
| 1 | 824     | hatchery | 3   | 0       |
| 2 | 7195    | hatchery | 4   | 0       |
| 3 | 8234    | natural  | 2   | 0       |
| 4 | 5901    | natural  | 3   | 0       |
| 5 | 130     | hatchery | 3   | 0       |
| 6 | 753     | hatchery | 3   | 0       |

The second output is the `hatchery_ages` data frame, with estimates of the total abundance of hatchery-origin fish for each age-class, for each iteration of the  $k$ -draw process. For each tagged fish recovered, the  $k \sim NB(1, \theta)$  draw to provides an estimate of the number of tagged fish not recovered. This implements step 1 of the CR methodology described in Equation 1 .

The repetitive iterations help characterize the uncertainty of our hatchery-origin estimate ( $\hat{N}_{CWT,a}$ ) due to  $\theta$ . The table has number of records equal to the number of age-classes  $a$  times the number of  $k$  iterations, and contains the following fields:

- `age` = target age-class of hatchery-origin fish.
- `N` = the number of recovered fish with tags for each age-class.
- `total_tags` = the estimate of the total number of fish with tags in the population for that age-class.  $k + 1$  is the estimate of the number of tagged fish available for each tag recovered Equation 1.
- `total_hatchery` ( $\hat{N}_{ho,a}$ ) = is the estimate of total abundance of hatchery-origin fish for each age-class in the population, estimated as  $\hat{N}_{CWT,a} * Fprod$  .
- `k_iteration` = is the number of the  $k$ -draw iteration.



```
#show table of estimates of hatchery origin fish
head(spawning_recoveries$hatchery_ages)
```

|   | age | N   | total_tags | total_hatchery | k_iteration |
|---|-----|-----|------------|----------------|-------------|
| 1 | 3   | 208 | 897        | 3588           | 1           |
| 2 | 4   | 112 | 667        | 2668           | 1           |
| 3 | 2   | 42  | 203        | 812            | 1           |
| 4 | 3   | 208 | 1012       | 4048           | 2           |
| 5 | 4   | 112 | 607        | 2428           | 2           |
| 6 | 2   | 42  | 282        | 1128           | 2           |

For a given age-class, the variation in **total\_hatchery** across **k\_iterations** reflects the uncertainty introduced by incomplete sampling and tagging rates.

### Simulate scale sampling and aging

We can now pass the fish “recovered” and our estimates of hatchery-origin abundance for each age-class onto the workhorse function **sim\_scales\_sampling.R**. This function simulates sampling of scales from recovered fish, bootstraps scale samples to characterize uncertainty, and produces estimates of natural-origin abundance for each age-class, along with associated uncertainty metrics. The inputs for **sim\_scale\_sampling()** are:

- **pop** = the simulated population data frame produced by **sim\_pop()**.
- **spawning\_results** = the output of the **sim\_spawning\_recovery()** function. Must be a list with two data frames: **spawning\_recoveries\$recovered\_fish** and **spawning\_recoveries\$hatchery\_ages**.
- **iterations** = number of draws from a negative-binomial distribution to estimate  $k$ , where  $k \sim NB(1, \theta)$ . Default 1000. In this function also the number of iterations to bootstrap resampling of the scale subsample to characterize scale sampling uncertainty.
- **scales\_n** = number of scale samples to collect and “age” from recovered fish. For this example we simulate collecting 500 scales.
- **CI** = confidence level used to calculate upper and lower confidence intervals of natural-origin age-class estimates produced from scale **iterations** bootstrapping. Default 0.95.

Running the function below with the default values and our simulated population and recoveries typically takes between 15 - 30 seconds on a standard workstation. It’s useful to time how long this process can take as it will be repeated many times over different scenarios later in the analysis.

```

sim_start <- Sys.time()
scale_results<-sim_scales_sampling(pop = sim1,
                                   spawning_results = spawning_recoveries,
                                   scales_n = 500)

sim_end <- Sys.time()
print(sim_end - sim_start)

```

Time difference of 14.2257 secs

The `sim_scales_sampling()` function starts by selecting a random sample of untagged fish records from the `spawning_recoveries$recovered_fish` input as our “aged scales”, assuming we know the true age of each fish in the sample, but not its origin. Then for each of the bootstrap iterations the `sim_scales_sampling()` function does the following:

1. Resample the scales from untagged recoveries with replacement.
2. Pull an iteration of the hatchery-origin age-class abundance estimates from the `spawning_recoveries$hatchery_ages` data frame.
3. Estimate the total number of fish without tags in population as  $\hat{N}_{NO-CWT}$  Equation 2.
  - Note: for our simulations  $\hat{N}$  is a known value from our simulation. In real surveys this value is a point estimate from carcass surveys with unreported uncertainty.
4. For each cohort  $a$  in the boot iteration:
  - calculate the number of untagged fish scales from that cohort ( $n_{NO-CWT,a}$ ).
  - calculate the proportion of untagged fish scales from that cohort ( $p_{NO-CWT,a}$ ).
  - estimate the total number of untagged fish in the populations from that cohort  $\hat{N}_{NO-CWT,a}$  Equation 3 .
5. Merge the untagged age-class estimates with the hatchery-origin estimates by age-class.
6. Estimate the number of untagged hatchery-origin fish in the population as  $\hat{N}_{NO-CWT,ho,a}$  Equation 4 .
7. Estimate the abundance of natural-origin fish in the population at each age-class  $\hat{N}_{no,a}$  Equation 5 .
8. Estimate the proportions of natural-origin fish at each age-class ( $\hat{p}_{NO,a}$ ). This is the proportion of the natural-origin population in each age-class, so that:

$$\sum_{a=1}^j \hat{p}_{NO,a} = 1$$

The results from each iteration are then used to produce summary statistics: mean proportions and abundances of each age-class of natural- and hatchery-origin fish, standard deviations, and lower and upper confidence intervals. Finally the function extracts the true age-class proportions and abundances from the simulated population.

The `sim_scales_sampling()` function returns four outputs: the `est_summary_stats` table, the `true_summary_stats` table, the `scales_collected` value, and the `boot_results` list.

```
str(scale_results$est_summary_stats)
```

```
Classes 'data.table' and 'data.frame': 3 obs. of 11 variables:
```

```
$ age           : int  2 3 4
$ mean_count_hatchery : num  839 4157 2236
$ mean_count_natural  : num  566 1271 932
$ mean_proportion_natural: num  0.206 0.457 0.337
$ sd_proportion_natural : num  0.0586 0.0865 0.0793
$ se_proportion_natural : num  0.00185 0.00274 0.00251
$ lower_CI_prop      : num  0.0927 0.296 0.188
$ upper_CI_prop      : num  0.325 0.631 0.489
$ lower_CI_count     : num  258 689 457
$ upper_CI_count     : num  873 1868 1382
$ n_iterations       : int  1000 1000 1000
- attr(*, ".internal.selfref")=<externalptr>
```

The `est_summary_stats` table contains the following fields:

- `age` = target age-class of fish.
- `mean_count_hatchery` = the mean abundance of hatchery-origin fish for each age-class  $\hat{N}_{ho,a}$  across bootstrapping iterations.
- `mean_count_natural` = the mean of abundance of natural-origin fish for each age-class  $\hat{N}_{no,a}$  from bootstrapping iterations.
- `mean_proportion_natural` = the mean proportion of natural-origin fish at each age-class  $\hat{p}_{no,a}$ .
- `sd_proportion_natural` = the standard deviation of the proportion of natural-origin fish at each age-class.
- `se_proportion_natural` = the standard error of the proportion of natural-origin fish at each age-class.
- `lower_CI_prop` and `upper_CI_prop` = the upper and lower confidence intervals of our natural-origin age-class proportion estimates.

- `lower_CI_count` and `upper_CI_count` = the upper and lower confidence intervals of our natural-origin age-class count estimates.
- `n_iterations` = the number of bootstrapping iterations performed.

In some bootstrap iterations, Equation Equation 5 yields negative values for natural-origin abundance, particularly for rarer age classes. E. Chen et al. [1] also runs into this issue and truncates the estimates to zero. In this implementation, negative estimates are retained (not truncated) to avoid introducing upward bias, though this can inflate the apparent uncertainty.

```
str(scale_results$true_summary_stats)
```

```
tibble [3 x 5] (S3: tbl_df/tbl/data.frame)
 $ age                : num [1:3] 2 3 4
 $ true_count_hatchery : int [1:3] 934 4213 2348
 $ true_count_natural  : int [1:3] 414 1393 698
 $ true_proportion_hatchery: num [1:3] 0.125 0.562 0.313
 $ true_proportion_natural : num [1:3] 0.165 0.556 0.279
```

The `true_summary_stats` table above contains the true counts and proportional information for hatchery- and natural-origin fish from the simulated population, used for comparison.

Looking at Figure 1 we see that there is substantial uncertainty around our estimates of natural-origin age-class proportions and counts. In this example, the true values fall within the 95% confidence intervals for all age classes.

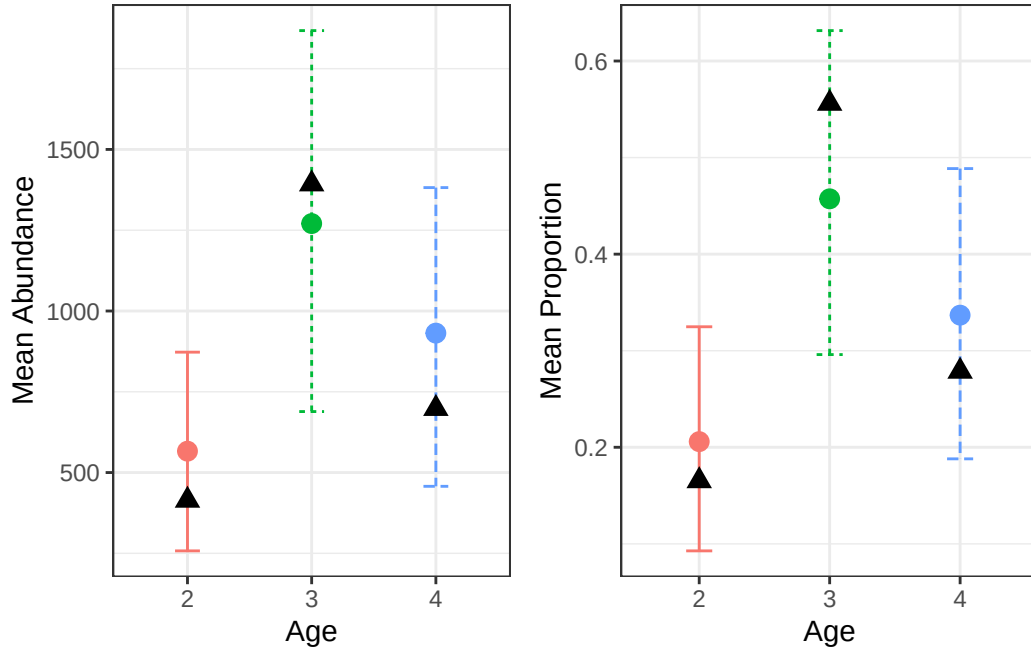


Figure 1: Abundance and proportional estimates with 95% confidence intervals for age-classes of natural-origin fish. True population values are shown with black triangles. Sample size of scales from untagged fish is 500.

Next we can use the `sim_summary_stats()` function to estimate the margin or error (MOE) for estimates of natural-origin counts and proportions for each age-class. This function reads the `scale_results` data created by `sim_scales_sampling()` and calculates:

Abundance MOE as:  $((\text{upper\_CI\_count} - \text{lower\_CI\_count}) / 2) / \text{mean\_count\_natural}$

Proportion MOE as:  $(\text{upper\_CI\_prop} - \text{lower\_CI\_prop}) / 2$

```
sim_stats<-sim_summary_stats(data=scale_results)
head(sim_stats)
```

|    | age   | count_bias | prop_bias   | count_moe | count_moe_relative | prop_moe  |
|----|-------|------------|-------------|-----------|--------------------|-----------|
|    | <num> | <num>      | <num>       | <num>     | <num>              | <num>     |
| 1: | 2     | 152.1764   | 0.04059757  | 307.5435  | 0.7428587          | 0.1160378 |
| 2: | 3     | -122.3644  | -0.09878023 | 589.4928  | 0.4231822          | 0.1677281 |
| 3: | 4     | 233.5440   | 0.05818266  | 462.2732  | 0.6622825          | 0.1502845 |

The `sim_stats` output table reveals some bias in the count and proportion estimates for each age-class. The relative MOE around our count estimates is quite high, particularly for the rarer

age classes (ages 2 and 4) while the MOE for the proportion estimates is substantially lower. this reflects the fact that count MOE scales with the magnitude of the estimate, while the proportion MOE is bounded between 0 and 1. This pattern of higher relative error for count estimates than proportion estimates persists across the simulations explored in the following sections.

### Bootstrapping simulations across scale sample sizes

We can run the above simulation through many iterations across a variety of scenarios. These are done in the `scripts/scenario_sims.R` script. Each scenario requires at least 2 hours of computation time for 100 bootstrap iterations. The results presented here are drawn from pre-computed outputs (`outputs/scale_sim_results.Rds`).

The first set of bootstrapping aims to see how the uncertainty in our estimate of age-class proportions of natural-origin fish changes across scale sample sizes Figure 2. The results show that as the scale sample size increases, the percentage of boot-strap iterations meeting a target MOE of 20% increases. **At a scale sample size of 900, roughly 95% of iterations achieve a MOE for the  $\hat{p}_{no,a}$  estimate below 20%.** It should be noted that none of the simulations produced a relative count MOE under 20% within the range of sample sizes tested. This will be addressed further in Approach 2.

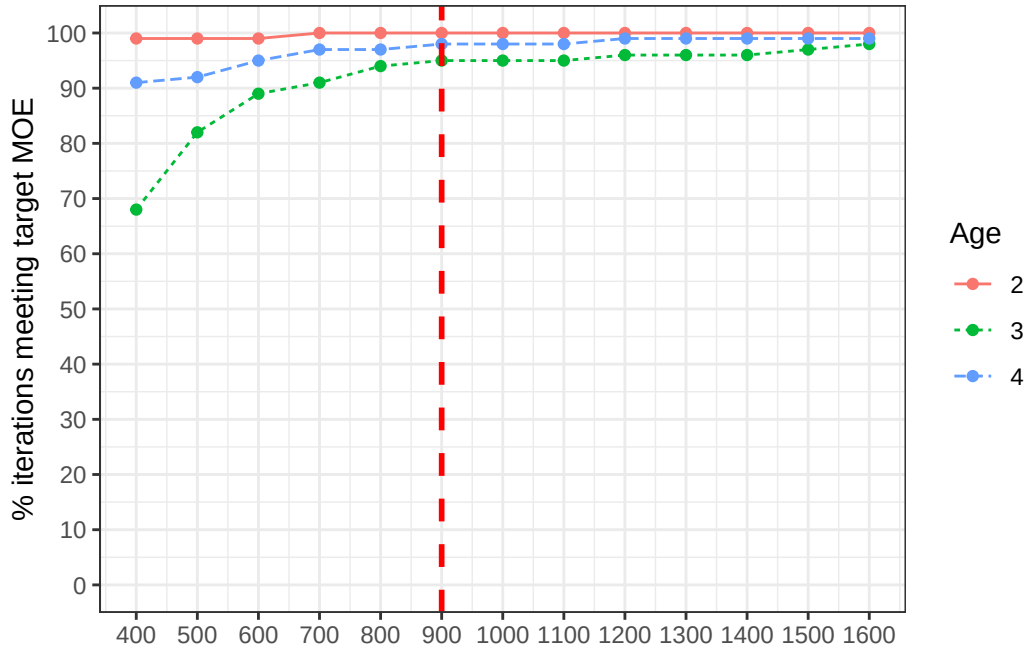


Figure 2: Percentage of simulation iterations that met target MOE of 20% for natural-origin age-class proportion estimates by scale sample size. The dashed red line indicates the smallest sample size at which 95% of iterations meet the target MOE (900 scales).

We can also double check our simulations below with Figure 3 by verifying whether the true values in our iterations fall within our 95% confidence intervals. Based on Figure 3 in general the true values for  $N_{no,a}$  and  $p_{no,a}$  fell within the 95% confidence intervals. Note that for these scenarios only 100 iterations were run per scale sample size due to processing time; with a larger number of iterations, coverage would likely approach the nominal 95% level for all age classes.

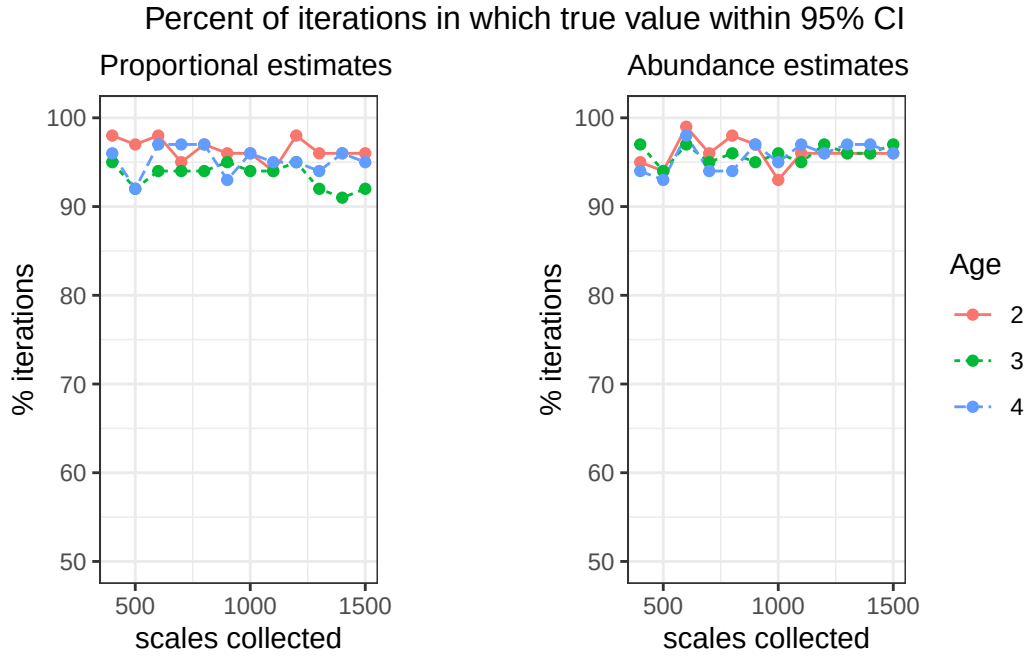


Figure 3: Percent of iterations at each scale sample size in which the true population value was within 95% confidence intervals.

I also ran a set of scenarios in which the scale sample size was fixed at 500, but the value for `theta`, or the sampling effort, increased. Figure 4 shows that at a `theta` of at least 0.4, there is 95% confidence that 500 scales sampled will give an estimate of the proportion of natural-origin fish in each age-class within the target MOE.

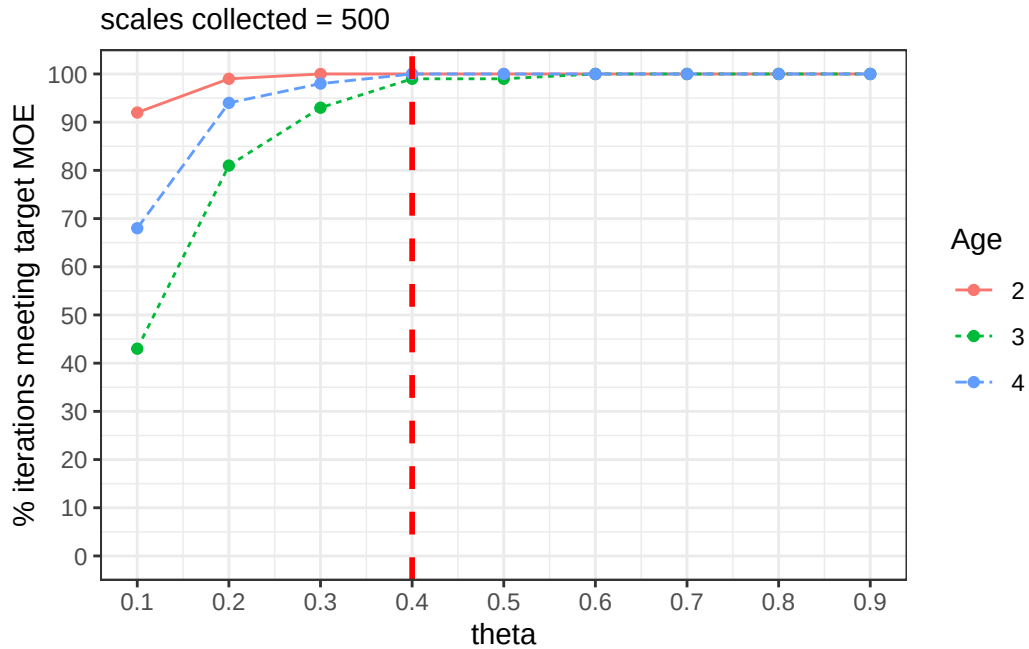


Figure 4: Percentage of simulation iterations that met target MOE of 20% by `theta`.



The next set of scenarios aimed to see how often rare abundances of natural-origin fish in a population resulted in “false negatives”, estimate of zero natural-origin fish in an age-class when they actually occurred. Figure 5 shows that false negative results were very rare. This is due to the k-draw method of estimating hatchery-origin abundances, which results in some natural fish always being estimated for each age-class, even if true abundances are very low.

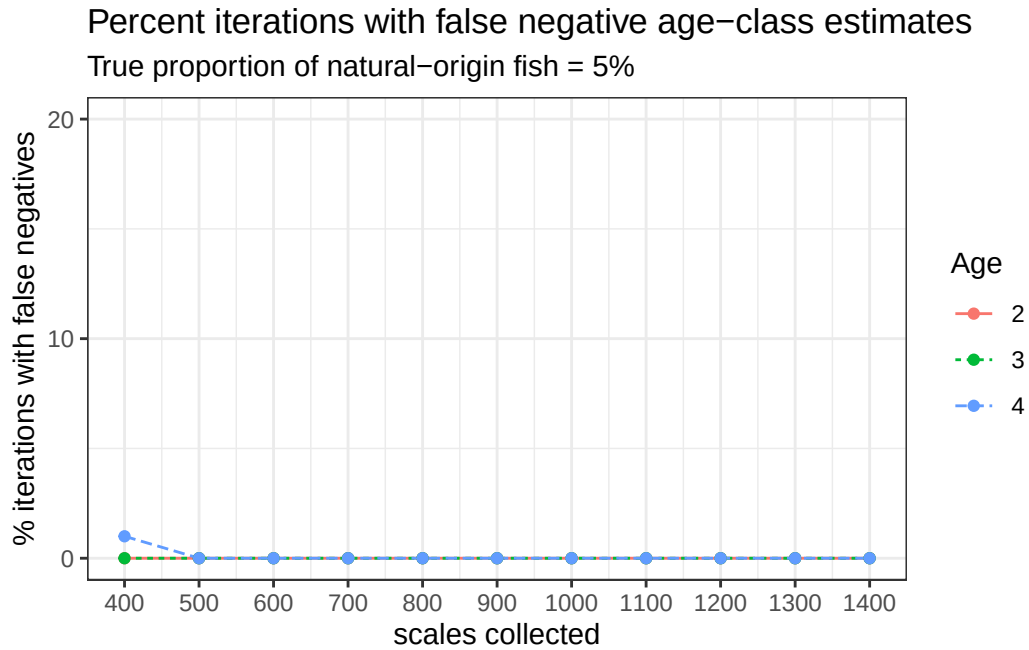


Figure 5: Percentage of simulation iterations that result in a false negative estimate of natural-origin age-classes. True proportion of natural-origin fish is 5%.

Figure 6 shows a different issue. In simulated populations where no natural-origin fish actually occurred, nearly every boot strap iteration resulted in estimates of natural origin fish where none occurred. This occurs because the  $k$ -draw method for estimating hatchery abundance introduces asymmetric uncertainty: when natural-origin fish are absent, the subtraction in Equation Equation 5 often yields positive estimates due to overestimation of hatchery-origin fish or underestimation of total untagged fish.

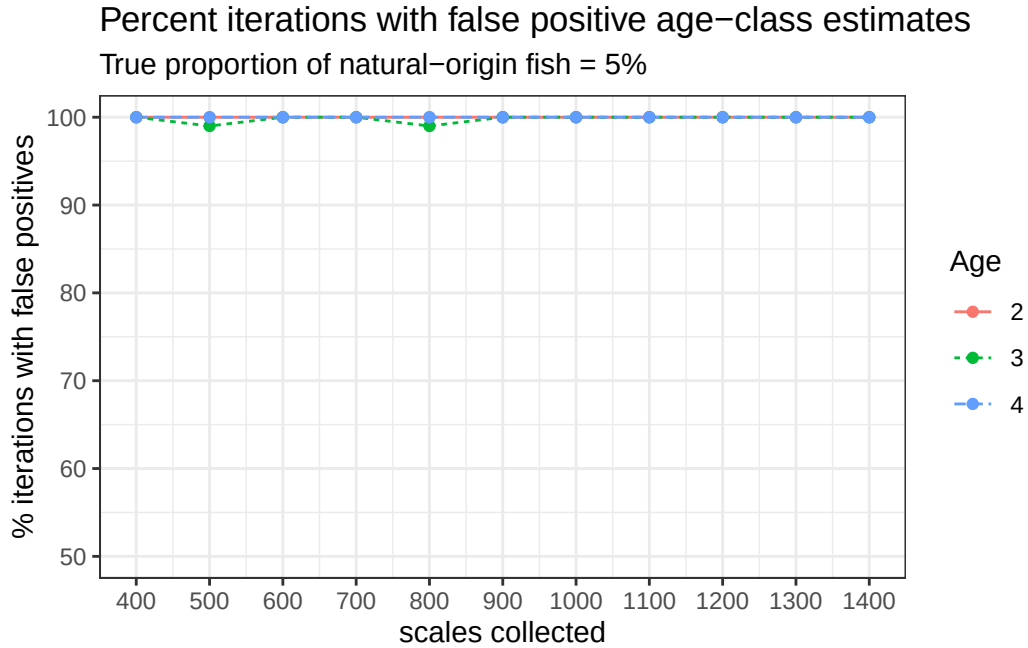


Figure 6: Percentage of simulation iterations that result in a false positive estimate of natural-origin age-classes.

The above simulations in Approach 1 set a base-line moving forward, showing that for a given simulated population, aging 900 scales from the untagged recovered fish will provide estimates of the proportion of natural-origin fish in each age-class within a  $\pm 20\%$  MOE.

## Approach 2: Sacramento River basin populations

In the previous section I demonstrated how the cohort reconstruction method works when estimating age-class proportions for natural-origin fish. This was a useful exercise to understand how the CR estimation methods are impacted by changes in scale sample size, sampling effort ( $\theta$ ), and to observe the few false-negatives but many false-positives. It also demonstrates the feasibility and utilization of the simulation functions. However, this was a theoretical and simplified system. To estimate basin-wide rates of maturation, mortality, and exploitation, we must simulate a sampling process that draws scales from various spawning grounds and hatcheries across the basin. The next step is to simulate a Sacramento River basin population that incorporates surveys of different tributaries and hatcheries and uses their sampling results to estimate the abundance of natural-origin fish at each age-class for the entire basin.

### Simulate populations from real data

First, we need to simulate our different populations across the basin. This time, instead of using theoretical numbers, we use real estimates and their related uncertainty to better simulate real world variation and sampling issues.

The `sim_basin_pop()` function simulates a population by taking random draws from the distribution of age-class abundance estimates produced by Emily Chen's work (<https://github.com/echenfishbitch/SRFC-cohort-reconstruction-wNF>). It requires the following inputs:

- `data` = a cleaned and formatted data set of survey based estimates containing distributions of age-class abundance estimates for each spawning ground or hatchery and each year, as well as tag rates. Data is formatted in the `clean_recovery_estimates.R` script to transform from brood year estimates to age-class estimates.
- `iter` = index of which estimate iteration to pull from, this is used when bootstrapping across available data for uncertainty.
- `year` = the target year for which population data will be pulled.

Before running an example of `sim_basin_pop()`, we can examine the input abundance estimate data here:

```
trib_data <- readRDS("data/all_returns_clean.Rds")
head(trib_data)
```

|    | iteration | return_year | location | count | age   | tag_rate  | origin   |
|----|-----------|-------------|----------|-------|-------|-----------|----------|
|    | <int>     | <num>       | <char>   | <num> | <num> | <num>     | <char>   |
| 1: | 1         | 2010        | American | 993   | 2     | 0.2648087 | hatchery |
| 2: | 1         | 2010        | Other    | 0     | 2     | 0.2500000 | hatchery |
| 3: | 1         | 2010        | Clear    | 76    | 2     | 0.2488500 | hatchery |

```

4:      1      2010 Feather  4138      2 0.2535023 hatchery
5:      1      2010      Sac   229      2 0.2491655 hatchery
6:      1      2010      Yuba   653      2 0.2481522 hatchery

```

The `trib_data` table contains 1000 iterations for each `return_year` and `location` combination providing estimates of the count of each age and origin class for returning fish.

We can run `trib_data` through our `sim_basin_pop()` function. The output is similar to the output of `sim_pop()` but this time includes the location of the fish and simulates the population of the entire basin pooled across different survey areas.

```

sim2 <- sim_basin_pop(trib_data,
                      iter = 1,
                      year = 2012)
head(sim2$iter_pop)

```

|    | return_year | location | age   | origin   | year  | has_cwt | fish_id |
|----|-------------|----------|-------|----------|-------|---------|---------|
|    | <num>       | <char>   | <num> | <char>   | <num> | <num>   | <int>   |
| 1: | 2012        | American | 2     | hatchery | 2012  | 0       | 1       |
| 2: | 2012        | American | 2     | hatchery | 2012  | 1       | 2       |
| 3: | 2012        | American | 2     | hatchery | 2012  | 0       | 3       |
| 4: | 2012        | American | 2     | hatchery | 2012  | 1       | 4       |
| 5: | 2012        | American | 2     | hatchery | 2012  | 1       | 5       |
| 6: | 2012        | American | 2     | hatchery | 2012  | 0       | 6       |

We also have the related  $\theta$  data for the historical hatchery and tributary survey data derived from the reported RMIS estimated number.

```

theta_data <- readRDS("data/location_year_theta.Rds")
head(theta_data)

```

```

# A tibble: 6 x 3
# Groups:   location [1]
  location return_year theta
  <chr>      <int> <dbl>
1 American    2008 0.193
2 American    2009 0.196
3 American    2010 0.187
4 American    2011 0.109
5 American    2012 0.161
6 American    2013 0.199

```

We can see in Figure 7 that the range of sampling effort is variable across the basin. Hatcheries tend to have very high  $\theta$  values, while many spawning grounds have much lower  $\theta$ . This will become important as we go forward, because locations with higher values for **theta** provide more recovered scales available for aging, and can bias final age-class estimates if not properly accounted for.

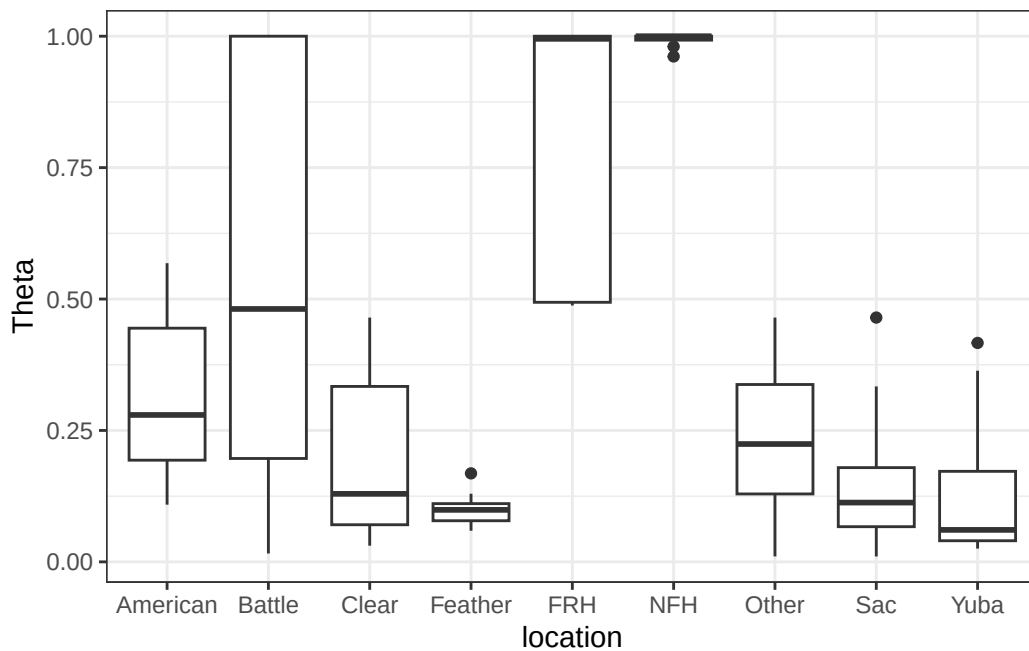


Figure 7: Theta values from 2010 to 2020 for spawning survey locations.

We can use the above **trib\_data** and **theta\_data** tables like we did with Approach 1, but now we must determine the best method for estimating abundances of natural-origin returns in each age-class for the entire Sacramento River basin.

There are two potential ways consider estimating basin wide abundances and proportions of natural-origin age-classes. The first involves aging a target number of scales collected from each spawning survey and hatchery, estimating age-class abundance and proportions for those locations, and then summing the results to get basin wide estimates. I'll call this **no-pooling** method. The second method involves collecting scales from all spawning surveys and hatcheries, and then randomly sampling a selection of the available scales from across the basin, aging that subsample, and then estimating basin-wide statistics using that selection. This will be called the **pooling** method.

### Estimating basin wide age-classes no-pooling

We can use similar methods to the earlier simulated theoretical population, but there are some significant differences so I'll walk through the approach in detail here. This section details the steps for the “no-pooling” approach, followed by a comparison with the “pooling” approach in the next section.

First, we select a year to draw population data from, and then run the `sim_basin_pop()` function to create our basin population. In this case I'll set the `target_year` to 2012, and set `iter` to 1 for the initial example. Below, we can create `pop`, the output of our population function, and `loc_list`, a list of all the locations sampled for scales.

```
target_year <- 2012

#draw population data
pop <- sim_basin_pop(trib_data,
                    iter = 1,
                    year = target_year)$iter_pop
#create list of survey locations
loc_list <- unique(pop$location)
loc_list
```

```
[1] "American" "Clear"      "Other"      "Feather"    "Sac"        "Yuba"        "FRH"
[8] "NFH"      "Battle"
```

We also want to grab the true abundance values for each natural-origin age-class across the basin, so we save them as `true_pop_ages`. These are used for comparison against our estimates.

```
true_pop_ages <- pop%>%
  filter(origin == "natural")%>%
  group_by(age)%>%
  summarise(true_abundance = n())
```

Next, simulate survey sampling fish recoveries at each location and the  $k$ -draw process for estimating hatchery-origin abundance with the `sim_spawning_recovery()` function. The below `for` loop iterates through each `location` and performs the following steps:

1. Pulls the true population data from `pop` for the target `location`.
2. Get's the relevant `theta` value for that `location` and `target_year` combination.
3. If there is no `theta` value, set's to a default of 0.2.
4. Runs the `location` and `target_theta` values through the `sim_spawning_recovery` function, and saves the output to the `recoveries_list`.

```

#spawning recoveries for each location
recoveries_list <- list()
for(l in 1:length(loc_list)){ #loop recoveries sim for each location
  iter_loc <- loc_list[l]
  loc_pop <- pop%>%filter(location == iter_loc) #get pop data for location

  target_theta <- theta_data%>% #get theta for location x year
    filter(location == iter_loc,
      return_year == target_year)

  if(nrow(target_theta) == 0)
    {target_theta = 0.2} #if no theta, set default
  else
    {target_theta <- round(target_theta$theta,2)}

  #get spawning recoveries
  spawning_recoveries <- sim_spawning_recovery(loc_pop,
                                              theta = target_theta,
                                              iterations = 1000)

  recoveries_list[[l]] <- spawning_recoveries
}
names(recoveries_list) <- loc_list

```

The above loop produces the `recoveries_list`, a list that contains the `sim_spawning_recovery()` output for each location surveyed in the population Section. For example, for the American River the output includes both the `recovered_fish` table and the estimate of `hatchery_ages`:

```
str(recoveries_list$American)
```

List of 2

```

$ recovered_fish:'data.frame': 9770 obs. of 7 variables:
 ..$ return_year: num [1:9770] 2012 2012 2012 2012 2012 ...
 ..$ location : chr [1:9770] "American" "American" "American" "American" ...
 ..$ age : num [1:9770] 3 2 3 4 3 3 4 3 3 3 ...
 ..$ origin : chr [1:9770] "natural" "natural" "hatchery" "hatchery" ...
 ..$ year : num [1:9770] 2012 2012 2012 2012 2012 ...
 ..$ has_cwt : num [1:9770] 0 0 1 1 0 0 0 0 0 0 ...
 ..$ fish_id : int [1:9770] 183382 141090 10294 95080 179362 179897 94740 19260 18305 14...
$ hatchery_ages:'data.frame': 3000 obs. of 5 variables:
 ..$ age : num [1:3000] 3 4 2 3 4 2 3 4 2 3 ...

```

```

..$ N          : int [1:3000] 873 70 69 873 70 69 873 70 69 873 ...
..$ total_tags  : int [1:3000] 5109 353 436 5348 419 524 5345 396 399 5758 ...
..$ total_hatchery: num [1:3000] 20436 1412 1744 21392 1676 ...
..$ k_iteration  : int [1:3000] 1 1 1 2 2 2 3 3 3 4 ...

```

Next, we can run another for loop over each location in the `loc_list` that runs the recovered fish from each location through the `sim_scale_sampling()` function and saves the outputs.

The below loop does the following for each location in the basin population:

1. Filter the `pop` data for the target location into the `loc_pop` value.
2. Pull the location recoveries and hatchery ages data from `recoveries_list`.
3. Run `sim_scales_sampling()` with location population and recoveries data. For these examples, I set `scales_n` to 500.
4. Transforms the `boot_results` of the scale aging using `pivot_wider()` into a table with age class abundance estimations of natural-origin spawners for each of the 1000 scale boot-strapping iterations.
5. Save `scales_collected`, the actual number of scales sampled and aged for that population. This is important because not all locations have enough sampling effort, or  $\theta$ , to recover enough fish each year to meet a target quota.
6. Save the outputs to the `location_results` list.

```

location_results <- list()
scales_collected <- list()
for(l in 1:length(loc_list)){
  target_loc <- loc_list[[l]]
  loc_pop <- pop %>% filter(location == target_loc)

  d <- recoveries_list[[l]]

  sim_result <- sim_scales_sampling(
    pop = loc_pop,
    spawning_results = d,
    iterations = 1000,
    scales_n = 500,
    CI = 0.95
  )

  scales_collected[[l]] <- sim_result$scales_collected
  l_results <- list_rbind(sim_result$boot_results) %>%

```



```

select(k_iteration, age, total_natural) %>%
pivot_wider(
  id_cols = k_iteration,
  names_from = age,
  values_from = total_natural
)
location_results[[1]] <- l_results
}

names(location_results) <- names(scales_collected) <- loc_list

```

The output of the `location_results` list for the American R. contains estimates of the abundance of each age-class for natural-origin fish for each  $k$ -draw iteration:

```
str(location_results$American)
```

```

tibble [1,000 x 4] (S3: tbl_df/tbl/data.frame)
 $ k_iteration: int  [1:1000] 1 2 3 4 5 6 7 8 9 10 ...
 $ 2          : num  [1:1000] 1671 1495 2867 2232 1680 ...
 $ 3          : num  [1:1000] 35094 33689 32187 31875 33004 ...
 $ 4          : num  [1:1000] 706 715 1448 660 635 ...

```

We can plot the outcomes for some of the different locations in Figure 8. If we look at the Yuba river, there are very few scales sampled and aged because the amount of untagged fish recovered was too low due to the very low  $\theta$ . The Yuba also had no actual natural-origin fish in the simulated population for this year, as well as no age 4 fish of any origin. This resulted in an accurate estimate of zero natural-origin age 4 fish, but significant bias in the estimates of age 2 and 3 natural-origin fish.

Nimbus Fish Hatchery (NFH) has a much higher  $\theta$  value, resulting in tight confidence intervals around the natural-origin escapement estimates. However, we see “false-positive” estimates of age 4 natural-origin spawners where there were none.

The Feather River scale aging resulted in very wide confidence intervals for our estimates driven by the very low  $\theta$  value. These location-level estimates show a challenge of the no-pooling approach: locations with low  $\theta$  produce more uncertain estimates, while locations with high  $\theta$  produce more precise but potentially biased estimates (false positives for age-4 on NFH). When summed across the basin, these location-specific errors can degrade overall accuracy.

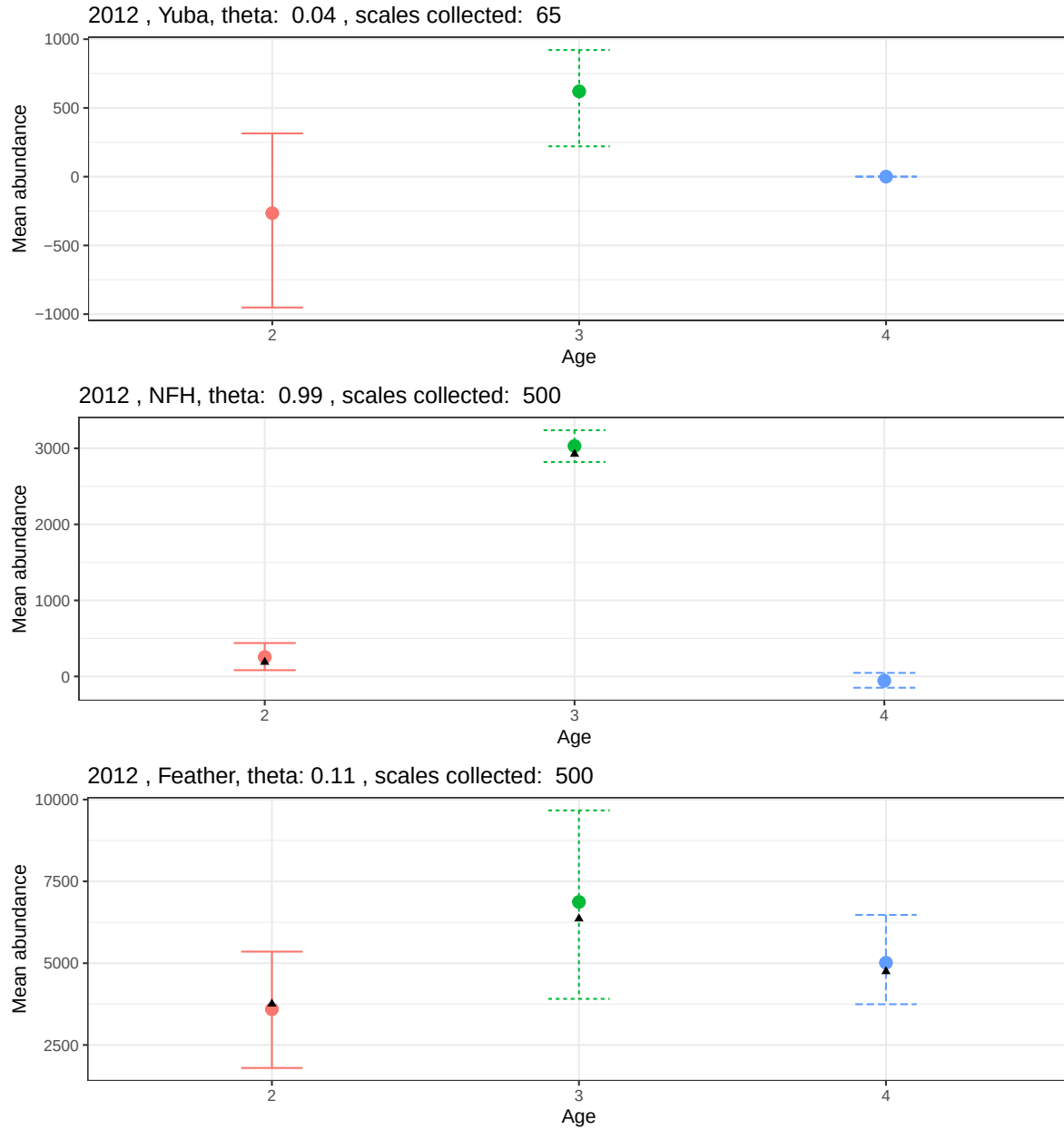


Figure 8: Abundance estimates with 95% confidence intervals for age-classes of natural-origin fish from three locations in the Sacramento River basin. Estimates are made without basin pooling. True population values are shown with black triangles.

We can sum the estimates of abundance from each location across each iteration to get estimates of total abundance for each natural-origin age-class in the basin population, and to characterize the uncertainty.

```
basin_results <- bind_rows(location_results, .id = "location") %>%
  group_by(k_iteration) %>%
  summarise(
    age_2 = sum(`2`, na.rm = TRUE),
    age_3 = sum(`3`, na.rm = TRUE),
    age_4 = sum(`4`, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  ungroup()
```

We can summarize the results of all the iterations to get the mean estimates and 95% confidence intervals for natural-origin abundance at each age-class for the entire basin.

```
CI = 0.95
basin_summary <- basin_results %>%
  pivot_longer(
    cols = starts_with("age_"),
    names_to = "age",
    values_to = "value"
  ) %>%
  group_by(age) %>%
  summarise(
    mean_natural_count = mean(value, na.rm = TRUE),
    lower_CI_count = quantile(value, (1 - CI) / 2, na.rm = TRUE),
    upper_CI_count = quantile(value, 1 - (1 - CI) / 2, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  ungroup()

basin_summary <- basin_summary %>%
  mutate(age = as.numeric(str_extract(age, "\\d+")))
```

Finally, we calculate how many scales we're sampled in total and join our estimates to the true population values to compare.

```
basin_summary$scales_collected <- sum(unlist(scales_collected))
basin_summary$target_location_scales <- 500
```

```
basin_summary <- basin_summary %>% left_join(true_pop_ages)
```

Figure 9 shows the final basin-wide estimates of natural abundance for each age-class, along with the 95% confidence intervals. The figure shows relatively tight intervals around our estimates, and the true values fall within these intervals for all age classes. However, this no-pooling method attempted to age 500 scales for each location, with a total amount of scales aged of 4065 out of a target basin-wide sample size of 4500. In this single example, the true values fall within the confidence intervals. However, as shown in the simulation results below, this is not consistently the case across years and sample sizes.

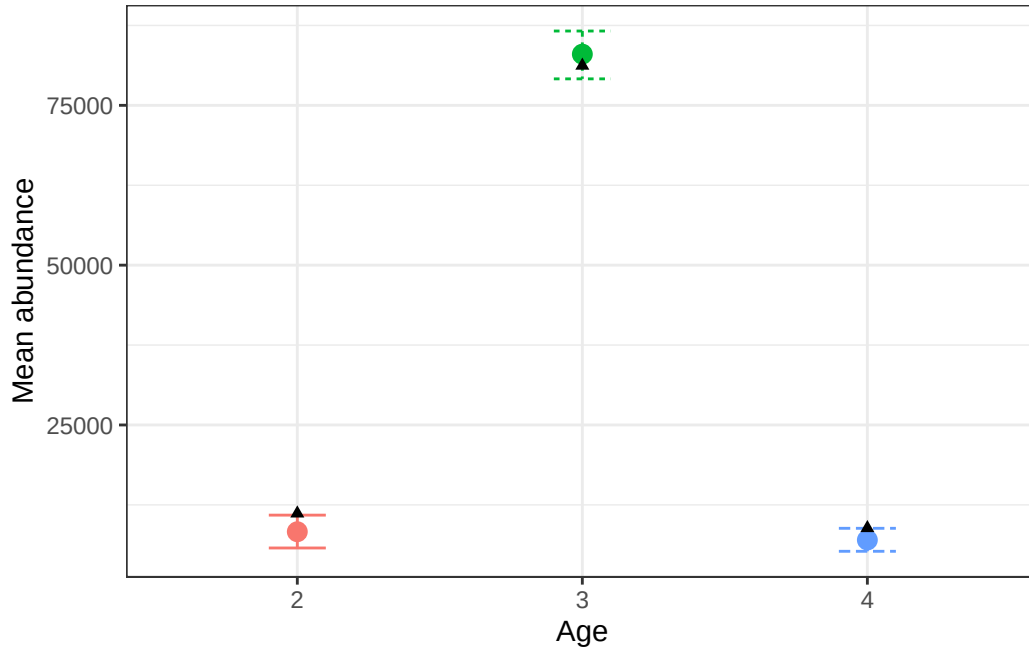


Figure 9: Abundance estimates with 95% confidence intervals for age-classes of natural-origin fish across the Sacramento River basin. Estimates are made without basin pooling. True population values are shown with black triangles.

Using this method, we can run a series of iterations (100 for these simulations), over a range of scale sample sizes and population years and see how the no-pooling method performs as we increase the sample size. The code for these Monte Carlo simulations can be found at `scripts/basin_sim.R` and the results are found at `outputs/basin_results_nopooling.Rds`.

Figure 10 shows a similar trend to the hypothetical populations simulated in Approach 1, but with very high relative MOE for the age 3 and 4 estimations even as the scale sample size gets

greater.

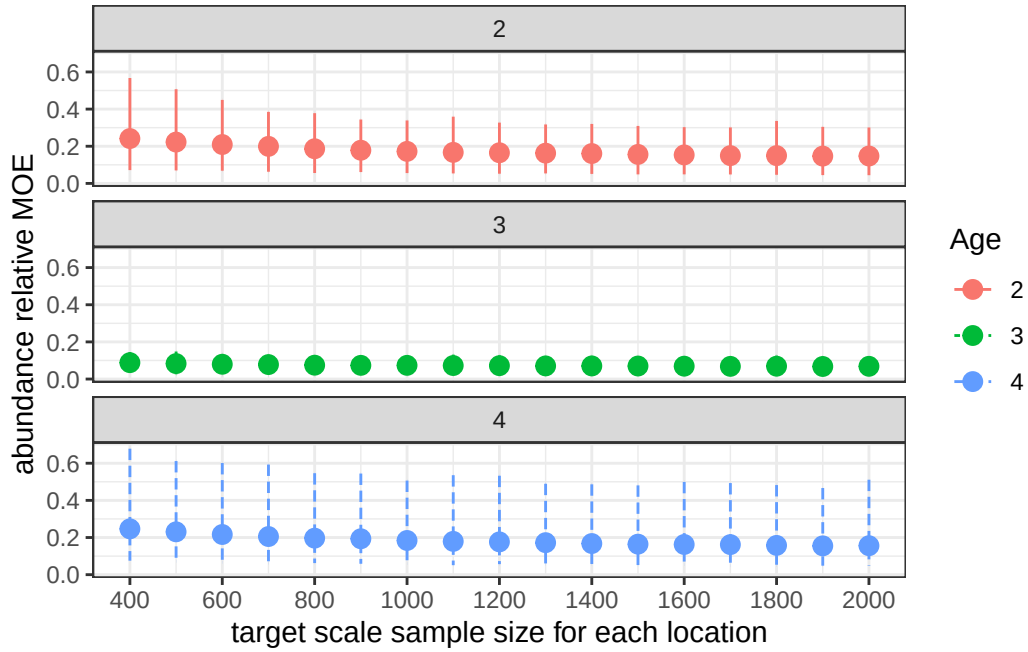


Figure 10: Basin-wide abundance estimate margin of error across target scale sample sizes for each age-class. Error bars represent 95% confidence intervals.

Figure 11 shows a trend that raises concerns about utilizing this method to estimate basin-wide age-class escapement. Many of the Monte Carlo iterations fail to include the true estimate of abundance for each age class within the lower and upper 95% confidence intervals and this value actually decreases as our scale sample sizes increase. Notably, coverage declines as scale sample size increases. This pattern of decreasing coverage despite larger sample sizes suggests that the no-pooling approach contains systemic bias that does not diminish with additional sampling.

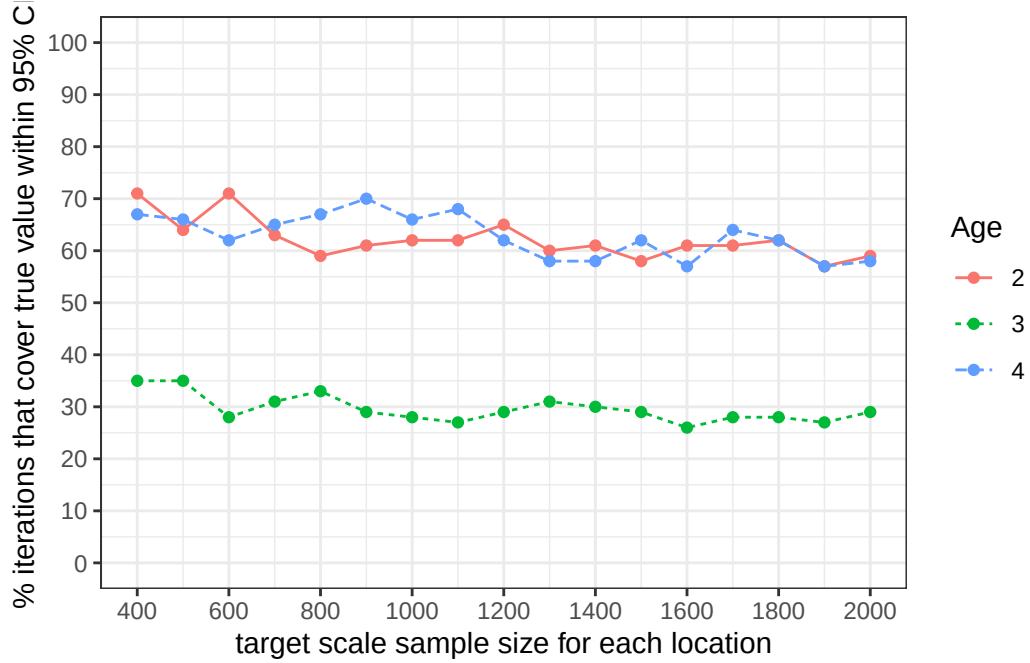


Figure 11: Percent of iterations at each scale sample size in which the true basin wide population value was within 95% confidence intervals.

Figure 12 shows that natural-origin abundance estimates for each age-class have some substantial bias, generally underestimating abundance in each age-class for all of our simulations. Since the bias does not change with sample size, this suggests there is some systemic bias in the sampling and estimation methods.

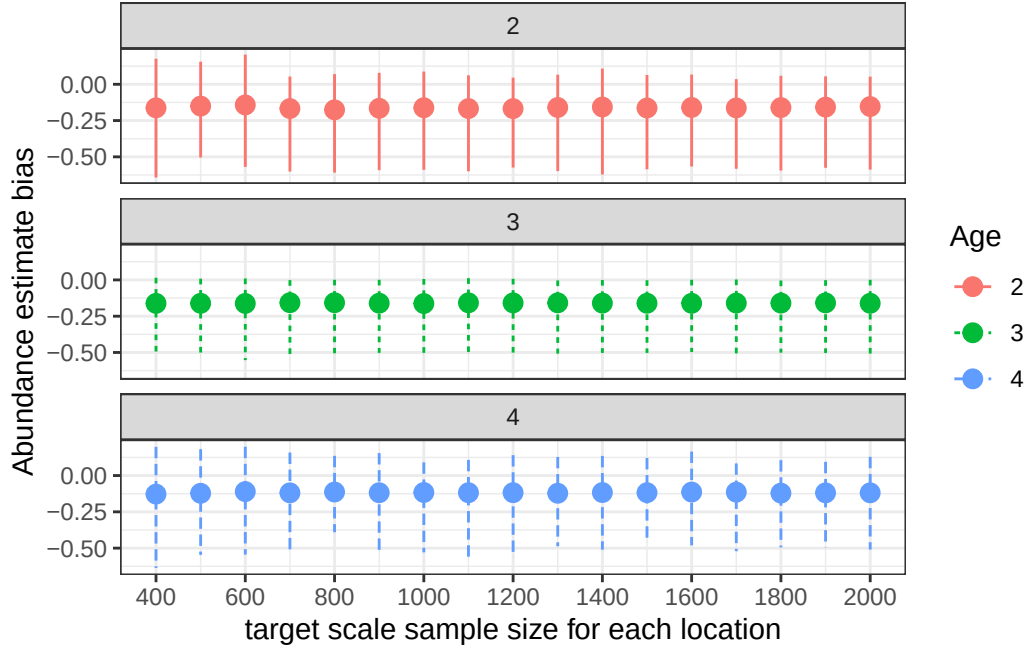


Figure 12: Relative bias in each age-class abundance estimate by target scale sample size for each location.

Reviewing Figure 13 reveals one likely cause of the bias. In 2012, locations like FRH and NFH (both hatcheries) have a higher proportion of recovered fish relative to their escapement contribution. This means that for a fixed target of 500 scales per location, hatcheries (with higher  $\theta$ ) can easily meet this target with high precision, while rivers like the American and Feather (with lower  $\theta$ ) may struggle to recover enough fish to reach the target. When location-specific estimates are summed to produce the basin-wide totals, estimates from hatcheries can receive disproportionate weight relative to their escapement contribution, while estimates from rivers can be based on fewer scales and carry greater uncertainty.

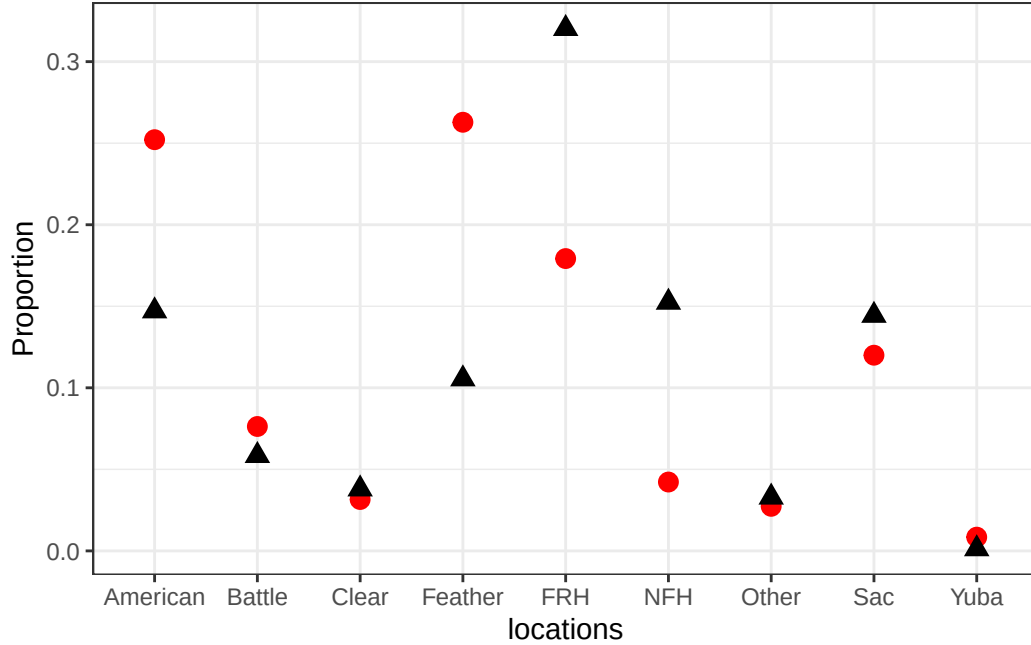


Figure 13: Proportion of escapement and recovered fish coming from each location. Red circles show the proportion of basin wide escapement represented by that survey location, black triangles show the proportion of basin wide fish recoveries available at that location. Basin population simulated by iter=1 and year=2012.

Another issue is illustrated in Figure 8, where estimates for the Yuba River age-2 and 3 fish are biased when there are no natural-origin fish present. In this example population, the bias of the Yuba river estimate can propagate to our basin-wide estimate when summing across locations, affecting the overall estimate of bias in estimates of age-2 natural-origin fish.

Both sources of bias originating from low- $\theta$  locations are limitations of the no-pooling approach. When we add more scale samples precision is added to locations that have plenty of samples available (hatcheries and spawning grounds with high  $\theta$ ), while low  $\theta$  locations remain poorly sampled, amplyfying this imbalance.



## Estimating basin wide age-classes with pooling

Using the basin wide simulated population `pop` already created, as well as the `recoveries_list` of all recovered fish and hatchery-origin estimates, we can replicate the above but this time by pooling our scale samples across locations. We'll start by making two new datasets from the `recoveries_list` for all locations in the simulated population. The code below combines all the recovered fish and hatchery abundance estimates into two dataframes, `all_recovered_fish` and `all_hatchery_estimates`.

```
all_recovered_fish <- all_hatchery_estimates <- data.frame()

for(l in 1:length(recoveries_list)){
  all_recovered_fish <- all_recovered_fish %>%
    rbind(recoveries_list[[l]]$recovered_fish)
  all_hatchery_estimates <- all_hatchery_estimates %>%
    rbind(recoveries_list[[l]]$hatchery_ages)
}
```

Next, we'll use the hatchery estimates from each location and for each  $k$ -draw to calculate estimates of basin wide hatchery abundance at each age-class for each iteration.

```
#sum for basin wide hatchery estimates for each k-draw iteration
setDT(all_hatchery_estimates)
basin_hatchery_estimates <- all_hatchery_estimates[, .(
  N = sum(N),
  total_tags = sum(total_tags),
  total_hatchery = sum(total_hatchery)
), by = .(age, k_iteration)]
```

Now we can address the weighting issue identified in the no-pooling approach. This is done by weighting the scale samples from each location by that locations proportion of total basin escapement, represented by `prop_esc`.

```
#get locations proportional escapement
basin_esc <- nrow(pop)
location_esc_prop <- pop %>%
  group_by(location) %>%
  summarize(esc = n(), .groups = "drop") %>%
  mutate(prop_esc = esc / basin_esc)
```

Next, we can use the `prop_esc` value for each location to weight the amount of scales to sample from that location. For this example we'll aim to age 1000 scales from across the entire basin.

Multiplying each location's `prop_esc` by the target scale sample total gives the number of scales (`target_scales`) to age from that location for the pooled basin estimates.

```
scales_n <- 1000
# use round_to_sum to get scales per location
location_esc <- location_esc_prop %>%
  mutate(target_scales = round_to_sum(prop_esc, scales_n))
head(location_esc)
```

```
# A tibble: 6 x 4
  location   esc prop_esc target_scales
  <chr>     <int>   <dbl>         <dbl>
1 American 61063   0.252           252
2 Battle   18474   0.0763            76
3 Clear     7631   0.0315            32
4 FRH      43408   0.179           179
5 Feather  63649   0.263           263
6 NFH      10224   0.0422            42
```

Now that we have the number of scales to sample from each location weighted by that locations contribution to basin-wide escapement, we can generate the list of scales for aging in the following `for` loop. For each location this loop:

1. Pulls the untagged available scales for that location.
2. If the number of available scales is greater than the number of `target_scales`, randomly sample them for aging and save to the `loc_recovered_scales` output data frame.
3. If the number of available scales is less than the `target_scales`, randomly sample what is available for aging and save to the output.

```
untagged_recoveries <- all_recovered_fish %>%
  filter(has_cwt == 0)

loc_recovered_scales <- data.frame()
for(l in 1:length(loc_list)){
  target_loc <- loc_list[l]
  ld <- location_esc %>% filter(location == target_loc)

  available_scales <- untagged_recoveries %>%
    filter(location == target_loc)

  if(nrow(available_scales) > ld$target_scales){
```

```

    # take a subsample of the available untagged fish
    loc_scales <- sample_n(available_scales, ld$target_scales, replace = FALSE)
  } else {
    loc_scales <- available_scales # take all available
  }
  loc_recovered_scales <- loc_recovered_scales %>% rbind(loc_scales)
}

```

The output `loc_recovered_scales` should be the same size as the number of scales targeted for the whole basin, 1000 in this example. In some simulations this may not be achievable as some locations do not recover enough untagged fish to fulfill the `target_scales` allocation. This shortfall is not redistributed to other locations to preserve the proportional representation of locations that can meet their targets.

```

basin_total_scales <- nrow(loc_recovered_scales)

basin_spawning_results<-list(
  "recovered_fish"=loc_recovered_scales,
  "hatchery_ages"=basin_hatchery_estimates)

```

The above code chunk combines the `loc_recovered_scales` with the `basin_hatchery_estimates` into the `basin_spawning_results` list, which is passed into the `sim_scales_sampling()` function below to get estimates of natural-origin abundance for each age-class for the entire basin.

```

basin_sim_result <- sim_scales_sampling(
  pop = pop,
  spawning_results = basin_spawning_results,
  iterations = 1000,
  scales_n = scales_n
)

basin_sim_result$target_basin_scales_n <- scales_n
basin_sim_result$actual_basin_scales_n <- basin_total_scales

```

Figure 14 shows the estimates of basin wide abundance and proportions for each age-class aging just 1000 scales pooled from across all locations. In this single simulation, estimates of abundance for each age-class show relatively tight confidence intervals and contain the true values within the upper and lower 95% confidence intervals.

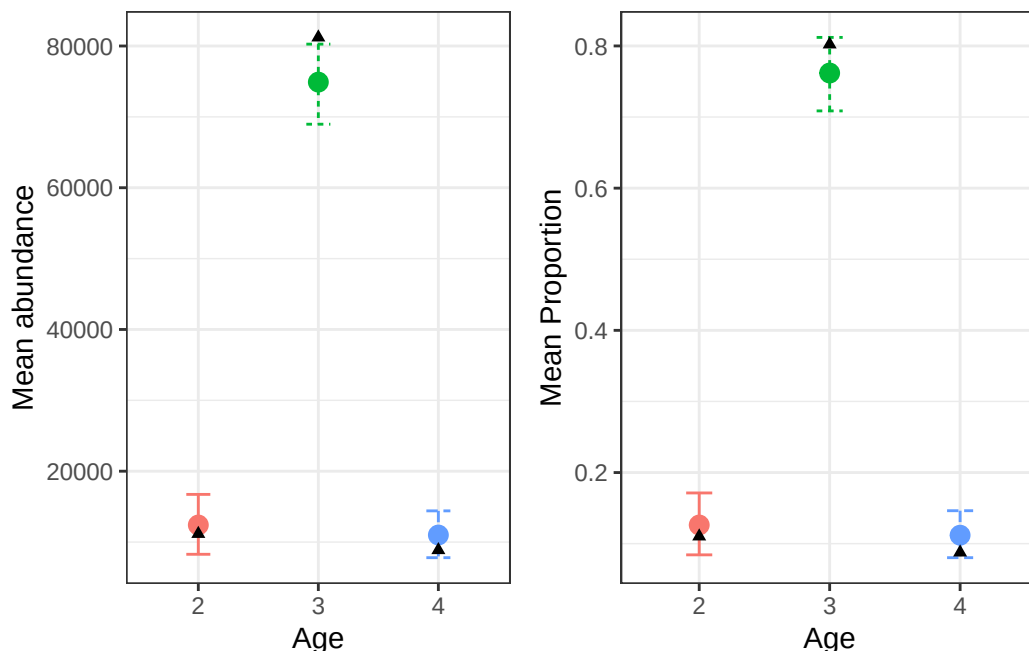


Figure 14: Abundance and proportional estimates with 95% confidence intervals for age-classes of natural-origin fish across the Sacramento River basin. Estimates are made by pooling scale samples from across all survey locations. True population values are shown with black triangles.

Just as with the previous approaches, we can run 100 iterations of the above Monte Carlo simulations for increasing scale sample sizes to examine how they impact the bias and MOE of the estimates. Figure 15 shows results of those simulations and how the MOE for both abundance estimates and age-class proportion estimates decreases with scale sample size. As we've seen in all the simulations so far, the margin of error for estimates of the proportion of each age-class of natural-origin fish is very low, below 10% for all the scale sample sizes. Relative MOE for abundance estimates is still high, especially for the rarer age-2 and 4 fish.

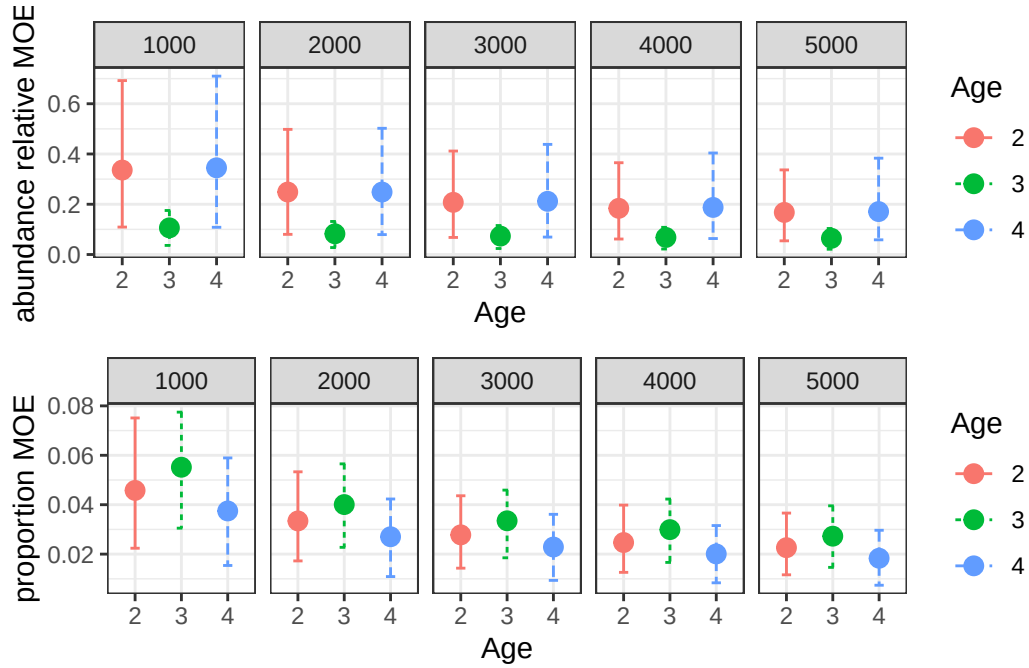


Figure 15: Abundance and proportional margin of error with 95% confidence intervals for age-classes of natural-origin fish across the Sacramento River basin iterated over increasing scale sample sizes for 100 Monte Carlo simulations.

Figure 16 shows that the scale sample pooling method results in coverage of the true population value at rates much higher than the no-pooling method. However, this coverage rate still decreases with higher scale sample sizes. This is because several of the locations with low recovery rates ( $\theta$ ) do not have enough scales available for aging, resulting in them being under represented in the basin as a whole when sample size targets exceed available fish. This shifts the effective weighting away from proportional escapement and toward high- $\theta$  locations, reintroducing some of the bias that the pooling approach was designed to mitigate.

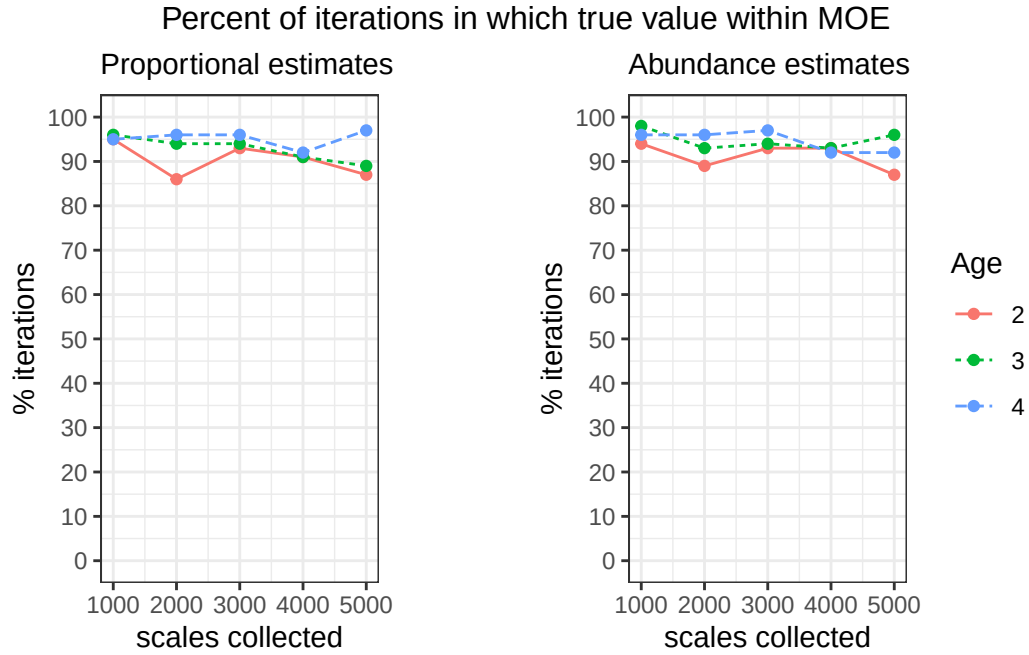


Figure 16: Percent of iterations at each scale sample size in which the true basin wide population value was within 95% confidence intervals.

The results of the Monte Carlo simulations demonstrate that the pooling approach outperforms the no-pooling approach. While both approaches produce similar patterns - decreasing MOE with increased sampling, higher uncertainty for rarer age-classes, and coverage decline at larger sample sizes - the pooling method gets much better coverage, lower bias, and is more efficient with scale aging effort (1000 total scales vs 4000+).

## Conclusions

Based on the result of these simulations I can address the questions put forth at the beginning of the document.

**Q1:** How many scales do we need to sample from recovered untagged fish to accurately estimate the proportion of age-classes in natural-origin fish in a given population?

**A:** Based on a hypothetical population with a fixed  $\theta$  of 0.2, at least 900 scales should be aged to ensure that the proportional estimates of each age-class have a MOE below 20%. However, this threshold varies with  $\theta$ ; higher sampling effort (larger  $\theta$ ) reduces the required sample size, while lower  $\theta$  increases it.

The simulations also show that false negatives, estimating an abundance of zero for an age-class when fish are present, happens at a negligible rate. In contrast, false positives occur at almost every instance in which natural-origin fish are absent but hatchery-origin fish are present. This is due to the uncertainty around the  $k$ -draw method for estimating hatchery-origin abundances.

**Q2:** Instead of aging some minimum number of scales from each tributary and hatchery, could we pool samples across the Sacramento River basin for a single CR? Could this be done to reduce effort/cost?

**A:** Sampling scales from a pool representing all locations is preferable to separately estimating age-class abundance for each location and summing across the entire basin. As long as the scales in the pool are weighted by the proportion of total escapement contribution from each location, this method results in the most representative estimates of basin-wide age-classes for natural origin fish.

The key assumption for this method is that the pool of available scales is representative of the basin-wide population of natural-origin fish. This assumption can break down at higher scale sample size as some spawning survey locations do not recover enough untagged fish to meet the weighted quota.

One key aspect lost through the pooling method is the ability to estimate age-class composition for each spawning location in the basin. While Cohort Reconstruction efforts are currently focused on estimating basin-wide rates, being able to estimate location specific rates, no matter how uncertain, could be useful in the future.

I suggest that we adopt the pooling method with location weighting based on proportional contribution to total escapement. This method minimizes the number of scales required for aging and is less biased than the alternative method.

## References

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- [3] Y Andre Wang and Mijke Rhemtulla. “Power Analysis for Parameter Estimation in Structural Equation Modeling: A Discussion and Tutorial”. In: *Advances in Methods and Practices in Psychological Science* (4.1 Jan. 2021).